College of Computer Science and Information Technology
Department of Computer Engineering

ARTI403: Image Processing

Lab 7(B): Mathematical and Statistical Filters

Student Name: Saad Abdulaziz Alshehri
Student ID: 2240006495

```
import cv2
import numpy as np

from IPython.display import Image, display
from matplotlib import pyplot as plt

plt.rc('text', usetex=False)

# ========== ( Adding nosie to an Image ) =================

# 0. Read an Image
imgPath = 'images/noise.jpg'
img = cv2.imread(imgPath, 0)

np.random.seed(42)  # same noise every run

# 1. Function that handle all noise algorithm (Guassian, Exp, Uniform & Salt&Pepper)
def add_noise(image, noise_type):
    row, col = image.shape

    if noise_type == "gaussian":
        mean = 0
        var = 0.01  # Adjust for more/less noise
        sigma = var**0.5
        gauss = np.random.normal(mean, sigma, (row, col))
        noisy = image / 255 + gauss
        return np.clip(noisy * 255, 0, 255).astype(np.uint8)

    elif noise_type == "salt_pepper":
        s_vs_p = 0.5
        amount = 0.04
        noisy = np.copy(image)
        # Salt (white pixels)
        num_salt = np.ceil(amount * image.size * s_vs_p)
        coords = [np.random.randint(0, i - 1, int(num_salt)) for i in image.shape]
        noisy[tuple(coords)] = 255
        # Pepper (black pixels)
        num_pepper = np.ceil(amount * image.size * (1.0 - s_vs_p))
        coords = [np.random.randint(0, i - 1, int(num_pepper)) for i in image.shape]
        noisy[tuple(coords)] = 0
        return noisy

    elif noise_type == "exponential":
        # Scale is the inverse of the rate (lambda)
        expo = np.random.exponential(scale=10.0, size=(row, col))
        noisy = image + expo
        return np.clip(noisy, 0, 255).astype(np.uint8)

    elif noise_type == "uniform":
        low = -20
        high = 20
        uni = np.random.uniform(low, high, (row, col))
        noisy = image + uni
        return np.clip(noisy, 0, 255).astype(np.uint8)



# 2. Load image in grayscale
img = cv2.imread(imgPath, 0)

# 3. Generate Noisy Images
noise_gauss = add_noise(img, "gaussian")
noise_sp = add_noise(img, "salt_pepper")
```

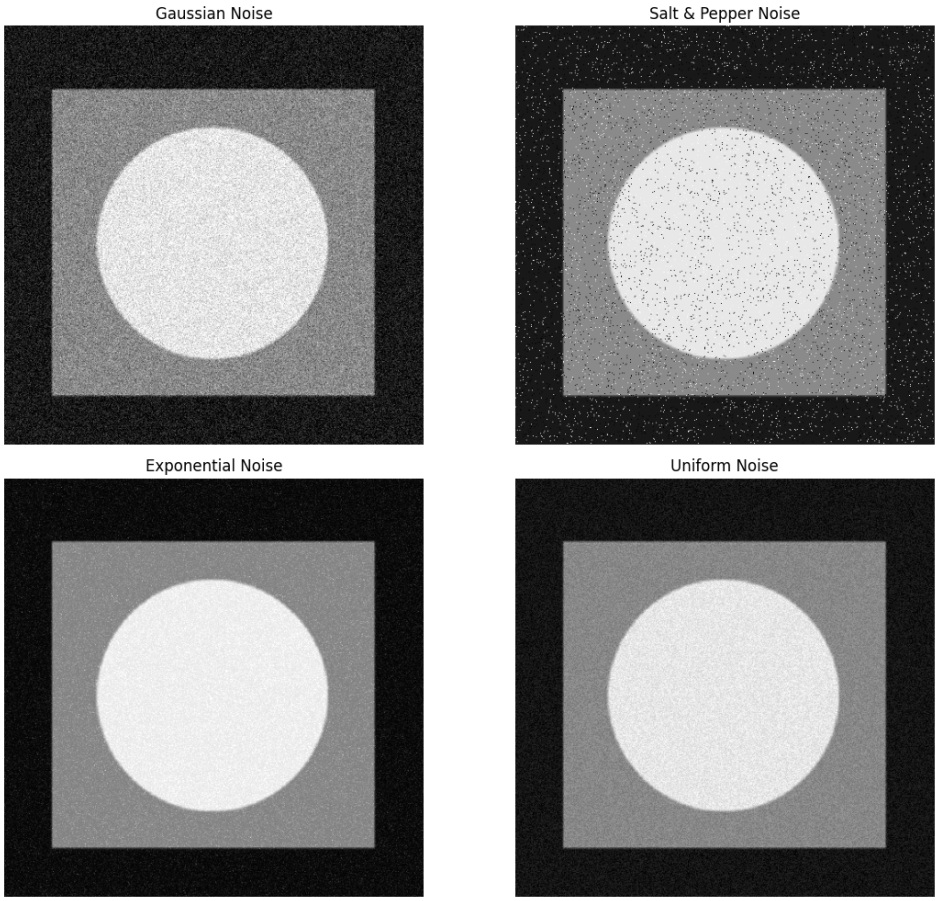
```
noise_expo = add_noise(img, "exponential")
noise_uni = add_noise(img, "uniform")

# 4. Display in 2x2 Grid
plt.figure(figsize=(12, 10))

plt.subplot(221), plt.imshow(noise_gauss, cmap='gray'), plt.title('Gaussian Noise')
plt.subplot(222), plt.imshow(noise_sp, cmap='gray'), plt.title('Salt & Pepper Noise')
plt.subplot(223), plt.imshow(noise_expo, cmap='gray'), plt.title('Exponential Noise')
plt.subplot(224), plt.imshow(noise_uni, cmap='gray'), plt.title('Uniform Noise')

for i in range(1, 5):
    plt.subplot(220 + i).axis('off')

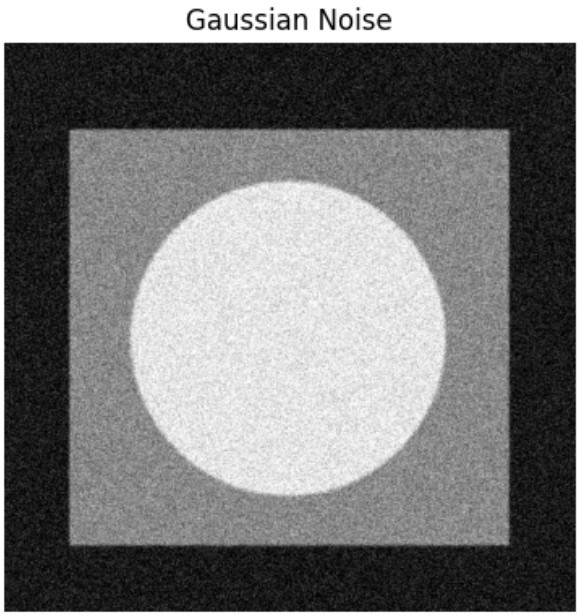
plt.tight_layout()
plt.show()
```

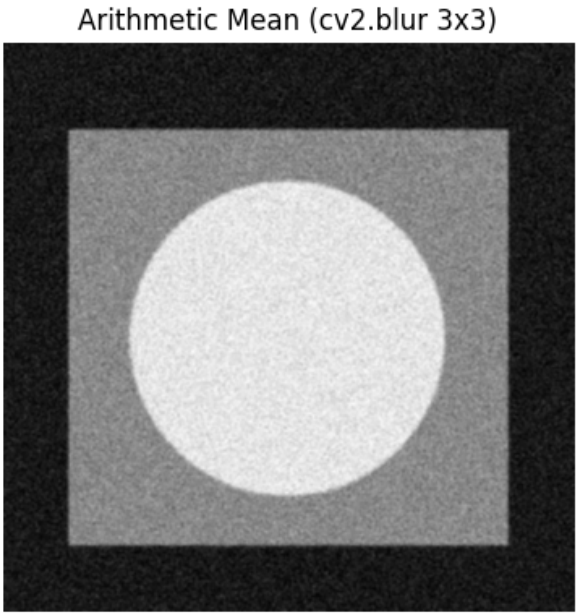



```
# --- METHOD 1: Using Built-in OpenCV function ---

# 2. Apply Arithmetic Mean Filter (using a 5x5 blur window)
# The larger the numbers in (3, 3), the smoother the image becomes.,
# you can change kernel size to (5,5), (7,7) etc.
filtered_img = cv2.blur(noise_gauss, (3, 3))

plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_gauss, cmap='gray'), plt.title('Gaussian Noise'), plt.axis('off')
plt.subplot(122), plt.imshow(filtered_img, cmap='gray'), plt.title('Arithmetic Mean (cv2.blur 3x3)'), plt.axis('off')
plt.show()
```

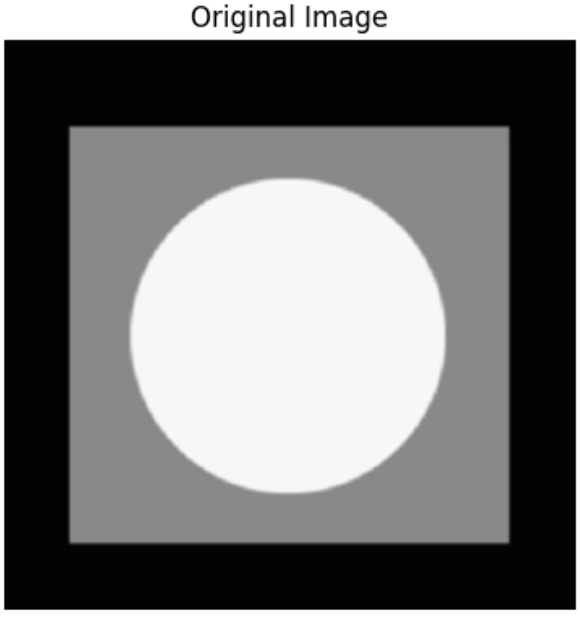


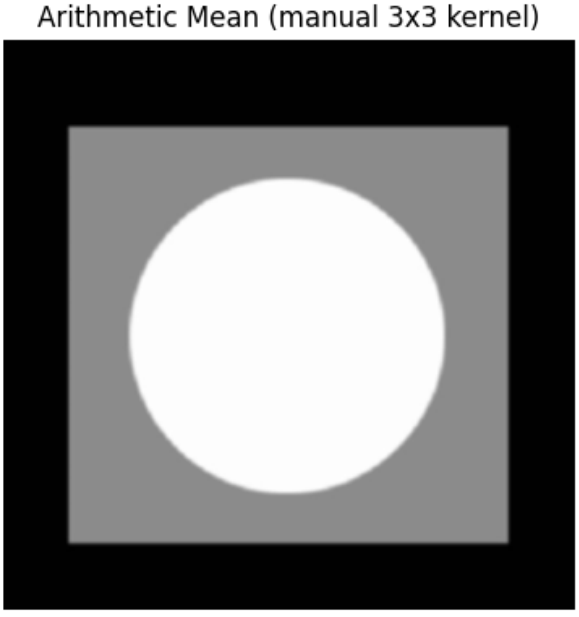



```
# --- METHOD 2: Manual implementation using a Kernel ---

# An Arithmetic Mean kernel is just a matrix of 1s divided by the total number of pixels.
kernel_size = 3
kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_size * kernel_size)
arithmetic_mean_manual = cv2.filter2D(img, -1, kernel)

plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('Original Image'), plt.axis('off')
plt.subplot(122), plt.imshow(arithmetic_mean_manual, cmap='gray'), plt.title('Arithmetic Mean (manual 3x3 kernel)'), p
('off')
plt.show()
```





Geometric Means

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def geometric_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 and add a tiny epsilon to avoid log(0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Apply the log-transform
    log_img = np.log(img_float)
```

```
    # 3. Apply a box filter to the log image (calculates the average of logs)
    kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_size * kernel_size)
    mean_log = cv2.filter2D(log_img, -1, kernel)

    # 4. Take the exponent to get back to the geometric mean
    geo_mean = np.exp(mean_log)

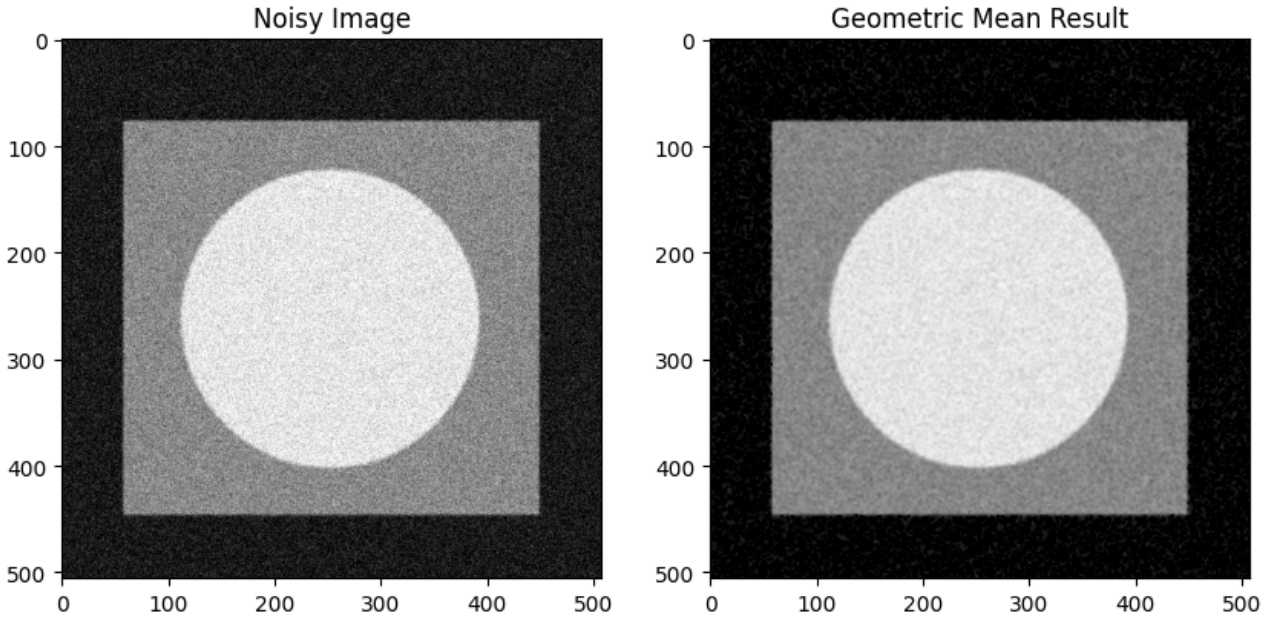
    # 5. Clip and convert back to 8-bit
    return np.clip(geo_mean, 0, 255).astype(np.uint8)


# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply filter
result = geometric_mean_filter(noise_img, kernel_size=3)

# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Image')
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('Geometric Mean Result')
plt.show()
```



Harmonic Means Filter

```
def harmonic_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 to avoid overflow during calculation
    # Add a tiny epsilon to handle black pixels (0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate the reciprocal of every pixel (1/P)
    reciprocal_img = 1.0 / img_float

    # 3. Sum the reciprocals in a window and divide by window size
    # This is equivalent to applying a standard mean filter to the reciprocal image
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    sum_reciprocal = cv2.filter2D(reciprocal_img, -1, kernel)

    # 4. Final step of the formula: (Window Size) / Sum of Reciprocals
    # Note: Window size is kernel_size * kernel_size
    n = kernel_size * kernel_size
    harmonic_mean = n / sum_reciprocal

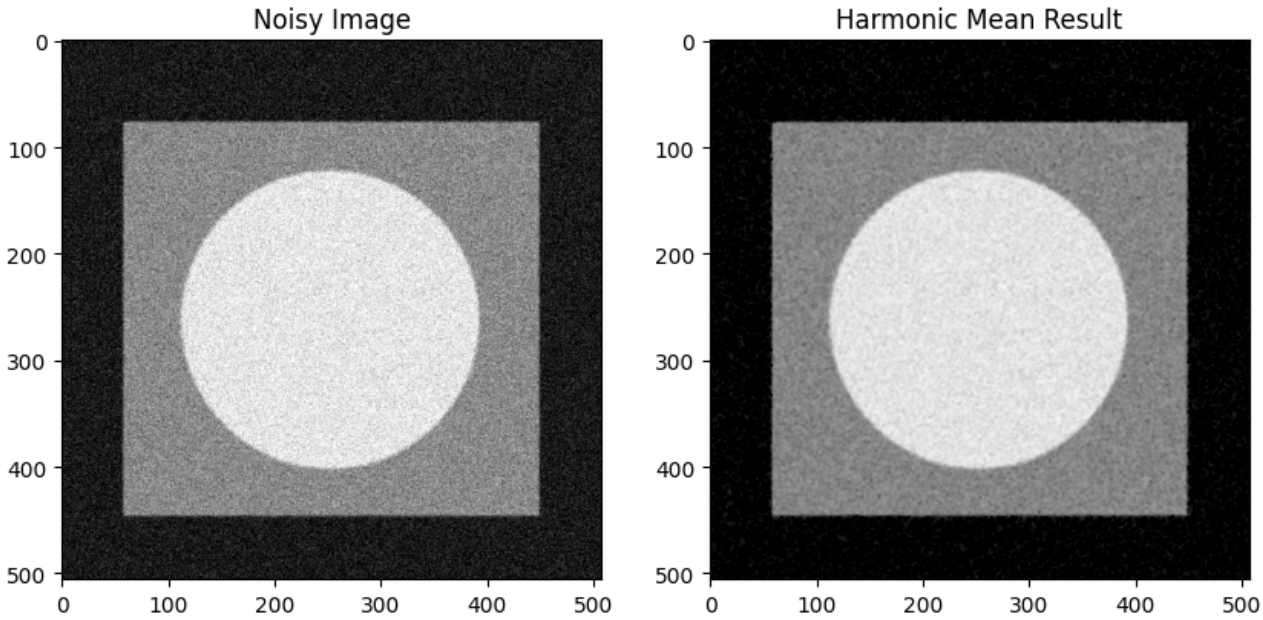
    # 5. Clip the results and convert back to 8-bit
    return np.clip(harmonic_mean, 0, 255).astype(np.uint8)
```

```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply the filter
cleaned_img = harmonic_mean_filter(noise_img, kernel_size=3)

# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Image')
plt.subplot(122), plt.imshow(cleaned_img, cmap='gray'), plt.title('Harmonic Mean Result')
plt.show()
```



Contra - Harmonic Mean Filter

```
def contra_harmonic_mean(image, kernel_size=3, Q=0):
    # 1. Convert to float to handle high powers and prevent overflow
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate numerator: Sum of pixels to the power (Q + 1)
    numerator = np.power(img_float, Q + 1)
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    num_sum = cv2.filter2D(numerator, -1, kernel)

    # 3. Calculate denominator: Sum of pixels to the power (Q)
    denominator = np.power(img_float, Q)
    den_sum = cv2.filter2D(denominator, -1, kernel)

    # 4. Divide and clean up
    # Handle potential division by zero
    result = num_sum / (den_sum + 1e-6)

    return np.clip(result, 0, 255).astype(np.uint8)

# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

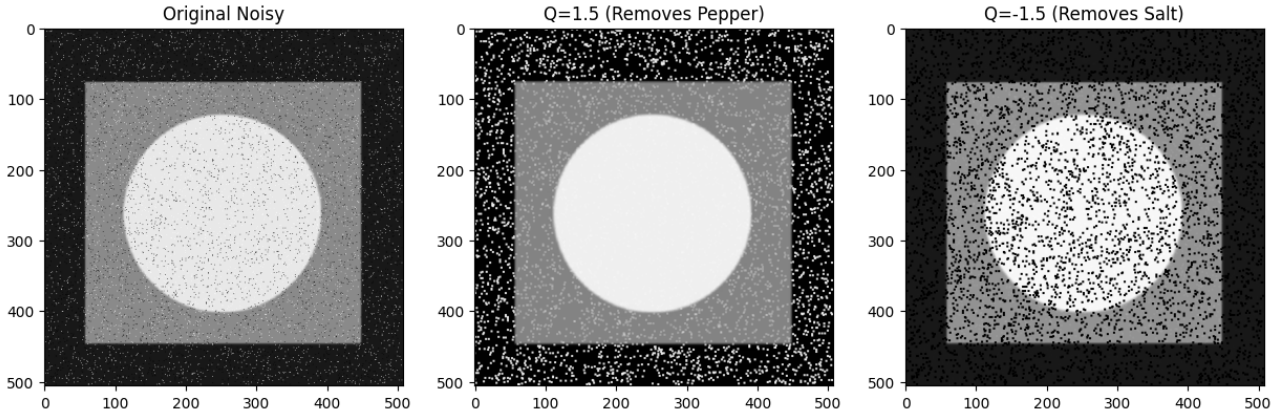
noise_img = noise_sp

# --- APPLICATIONS ---
# Q > 0 (e.g., 1.5): Removes Pepper Noise (black spots)
pepper_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=1.5)

# Q < 0 (e.g., -1.5): Removes Salt Noise (white spots)
```

```
salt_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=-1.5)

# --- DISPLAY ---
plt.figure(figsize=(15, 5))
plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original Noisy')
plt.subplot(132), plt.imshow(pepper_removed, cmap='gray'), plt.title('Q=1.5 (Removes Pepper)')
plt.subplot(133), plt.imshow(salt_removed, cmap='gray'), plt.title('Q=-1.5 (Removes Salt)')
plt.show()
```



Median Filter

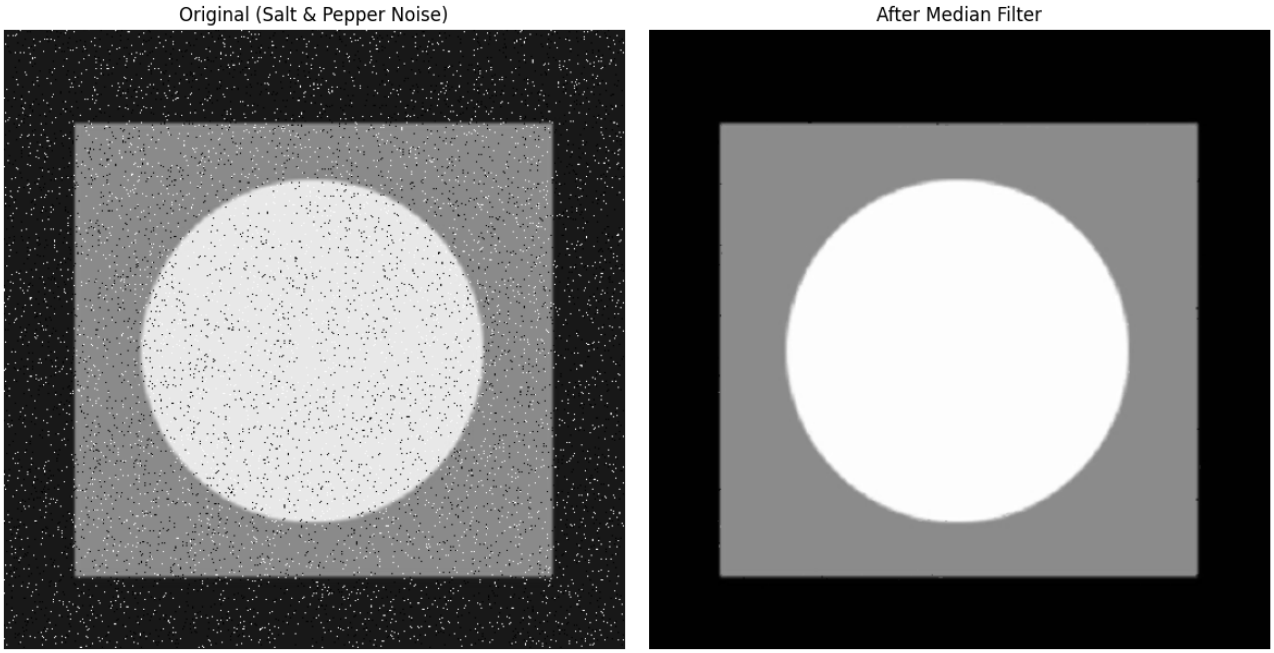
```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_sp

# 2. Apply Median Filter
# The second argument is the 'ksize' (kernel size).
# It MUST be an odd integer (3, 5, 7, etc.)
median_result = cv2.medianBlur(noise_img, 3)

# 3. Compare Results
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original (Salt & Pepper Noise)'), plt.axis('off')
plt.subplot(122), plt.imshow(median_result, cmap='gray'), plt.title('After Median Filter'), plt.axis('off')
plt.tight_layout()
plt.show()
```



Max and Min Filter

```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_sp



# 2. Define the Kernel size (e.g., 3x3)
kernel = np.ones((3, 3), np.uint8)

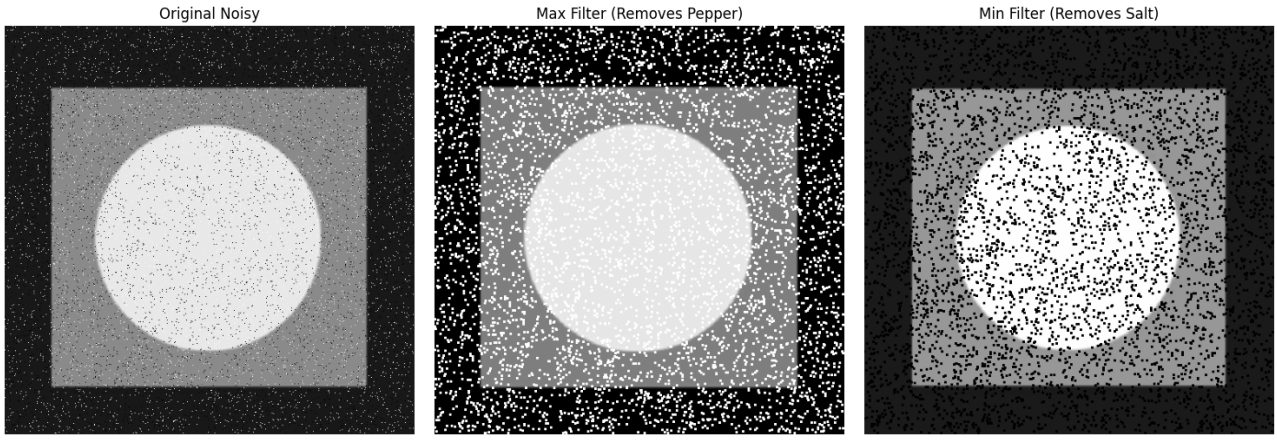
# 3. Apply Max Filter (Dilation)
# This picks the brightest pixel in the neighborhood.
max_filter = cv2.dilate(noise_img, kernel)

# 4. Apply Min Filter (Erosion)
# This picks the darkest pixel in the neighborhood.
min_filter = cv2.erode(noise_img, kernel)

# --- Display Results ---
plt.figure(figsize=(15, 5))

plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original Noisy')
plt.subplot(132), plt.imshow(max_filter, cmap='gray'), plt.title('Max Filter (Removes Pepper)')
plt.subplot(133), plt.imshow(min_filter, cmap='gray'), plt.title('Min Filter (Removes Salt)')

for ax in plt.gcf().axes: ax.axis('off')
plt.tight_layout()
plt.show()
```



Mid-Point Filtering

```
def midpoint_filter(image, kernel_size=3):
    # 1. Define the structural element (kernel)
    kernel = np.ones((kernel_size, kernel_size), np.uint8)

    # 2. Find the Maximum value in the neighborhood (Dilation)
    max_img = cv2.dilate(image, kernel).astype(np.float64)

    # 3. Find the Minimum value in the neighborhood (Erosion)
    min_img = cv2.erode(image, kernel).astype(np.float64)

    # 4. Calculate the average of Max and Min
    midpoint = (max_img + min_img) / 2

    # 5. Convert back to 8-bit
    return midpoint.astype(np.uint8)



# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
```

```
# noise_uni = add_noise(img, "uniform")

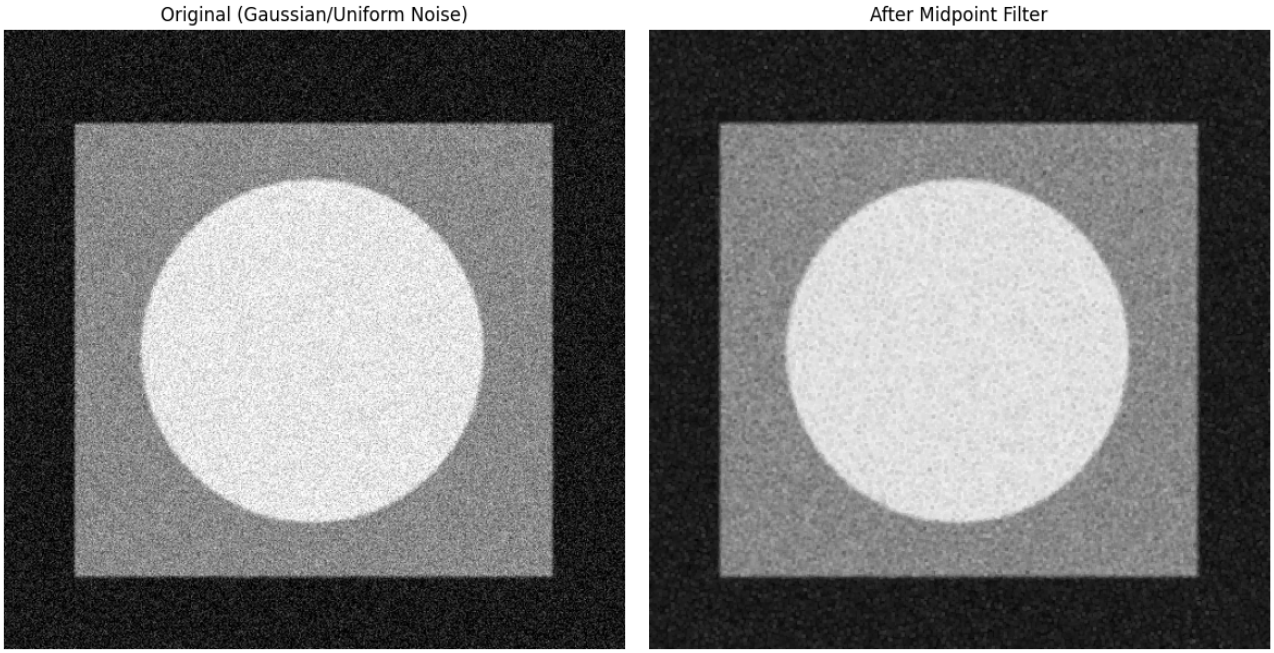
noise_img = noise_gauss


# Apply the Midpoint Filter
result = midpoint_filter(noise_img, kernel_size=3)

# --- Display Results ---
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original (Gaussian/Uniform Noise)'), plt.axis('off')
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('After Midpoint Filter'), plt.axis('off')

plt.tight_layout()
plt.show()
```

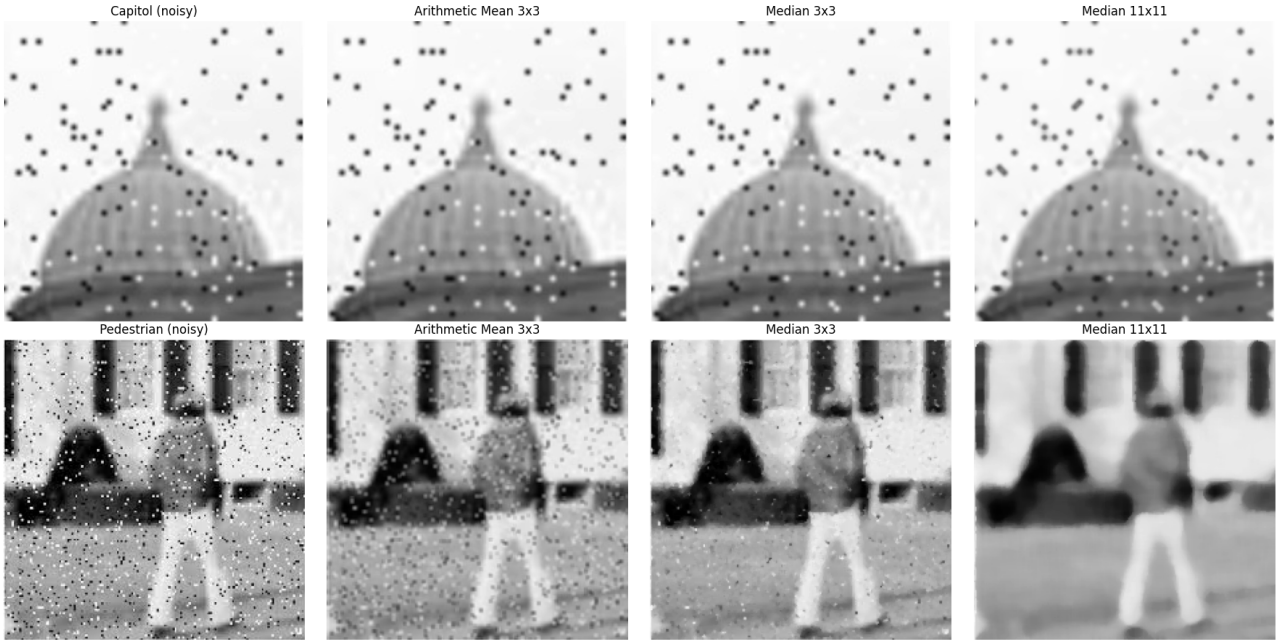


Removing noise from the given noisy images

The two images already have salt and pepper noise. I tried a 3x3 mean filter, a 3x3 median filter and an 11x11 median filter on both. The 3x3 median filter cleans the pedestrian image well. The Capitol image has big dots, so the filters only remove some of them.

```
capitol = cv2.imread('images/capitol_noisy.jpg', 0)
pedestrian = cv2.imread('images/pedestrian_noisy.jpg', 0)

fig, axes = plt.subplots(2, 4, figsize=(18, 9))
for row, (name, im) in enumerate([('Capitol', capitol), ('Pedestrian', pedestrian)]):
    results = [im, cv2.blur(im, (3, 3)), cv2.medianBlur(im, 3), cv2.medianBlur(im, 11)]
    titles = [f'{name} (noisy)', 'Arithmetic Mean 3x3', 'Median 3x3', 'Median 11x11']
    for ax, result, title in zip(axes[row], results, titles):
        ax.imshow(result, cmap='gray', vmin=0, vmax=255)
        ax.set_title(title)
        ax.axis('off')
plt.tight_layout()
plt.show()
```


Capitol (noisy)
Arithmetic Mean 3x3
Median 3x3
Median 11x11
Pedestrian (noisy)
Arithmetic Mean 3x3
Median 3x3
Median 11x11